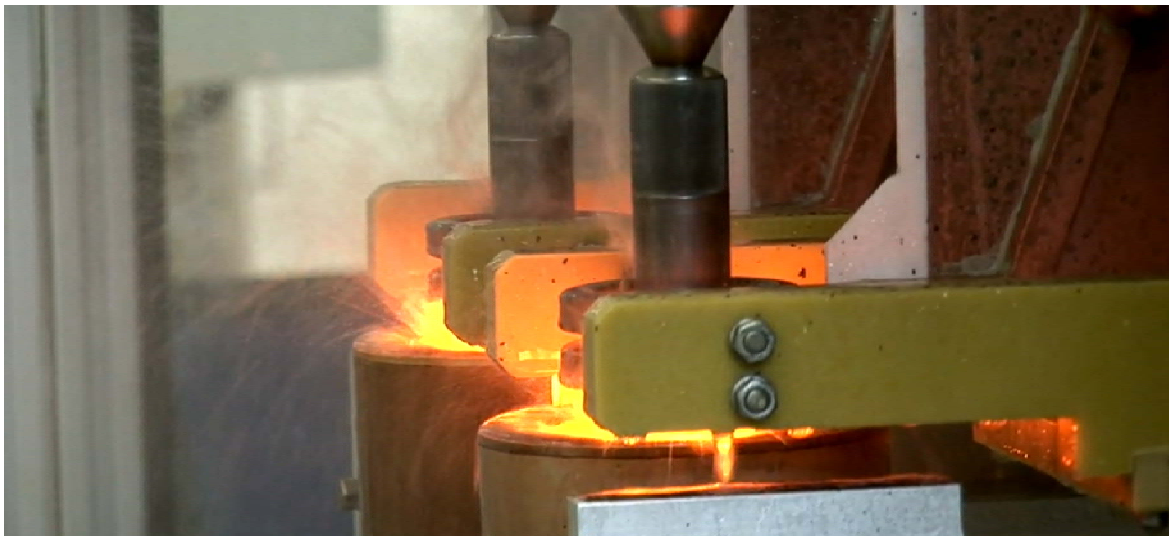


D/SHAFT INDUCTION HARDENING MACHINE



Customer:	HYUNDAI-WIA
Contact:	Mr. Taehon Lee
Inquiry No:	7002903.6
Workpiece:	Bar Shafts
Production Plant:	GH Electrotermia (Spain)
Date:	02/02/2017
Your contact in Sales Dept.	Pablo Arce
Phone:	+34 961352020 (ext 513)
Email:	parce@ghinduction.com

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1. Records

1.1. Introduction

7002903.6 version summarizes final commercial negotiation held on Changwon Hyundai WIA facilities on Feb 2nd.

7002903.5 is a last review including commercial conditions marked by Hyundai WIA for the project and adding an extraordinary price discount. Updated points have been marked in red for an easy and quick reading.

7002903.4 splits the prices according to Hyundai/WIA price list specifications chart.

This review 7002903.3 includes the update in the number of coils as well as excluding a common supply cabinet from supply and coming back to a multi-output power supply and independent output transformers set per station to have an independent control per unit.

Updated modifications have been marked in green for an easy and quick reading.

Review 7002903.2 includes the following updates on the previous quote:

- Simplified lay-out with one unique conveyor.
- Stations power all in series instead of independent.
- Modified mechanics.
- Service offer included.

Updated modifications on this review have been marked in orange for an easy and quick reading.

7002903.1 was the update following the previous quotation following the modifications on the last RFQ received on 8th of August and after the meeting hold in Hyundai-WIA Changwon on last July.

The update includes mainly the following modifications:

- All inductors referred to last part list are included on supply.
- Noise isolation up to <80dB is included as an option.

After checking all the drawings GH Induction confirms to accomplish all the given references with the same coil by scanning in the mentioned cycle time.

Flow of parts is just a reference as it will be defined after kick-off meeting. Lay-out will be sent attached on DWG file to this quotation.

Updated modifications have been marked in blue text for an easy and quick reading.

Following the request for quotation received from Hyundai-WIA for the supply of an induction hardening and tempering machine for bar shafts parts with final destination Monterrey (Mexico), and following the specifications of the document dated on ~~July 7th~~ August 8th 2016 signed by Mr. Hyun-Tae Park, please find hereafter our worked proposal.

The complete installation includes all the assemblies and devices required in the RFQ to follow completely its specifications for the treatment of D/Shfts parts.

GH Induction is proud to introduce HYUNDAI-WIA this proposal getting our best technical and service solution:

- Huge large experience on driveline heat treatment manufacturing with more than 300 references worldwide.
- High-tech exclusive solutions for induction and specifically driveline applications.
- Large experience in Mexico.

- GH Mexicana subsidiary company of GH Induction Group offers local technical service and quick intervention and spare parts supply.
- Fluent Spanish communication handled by our subsidiary in Mexico or our headquarters in Spain.

For all this and many more reason we have worked to make sure that GH Group offers to HYUNDAI-WIA the best and safest technical solutions. We remain fully available to describe, in detail, our offer and answer to any query you may have.

1.2. Project requirements

Part:	D/Shaft (Bar Shafts)
List of parts:	

☒ D/SHAFT specification data

NO	Size	Vehicle type	Item No	WORK data								Material	Remark
				A	B	C	D	E	F	G			
1	#18	D2UX	LH 28018-D2001	Ø 21	Ø 18.5	Ø 20.587	585.4	Ø 18.8	7	1.4		SAE10B38M2	
2			RH 28018-D2002	Ø 21	↑	↑	582.8	↑	↑	↑		↑	
3	#22	HE v1.8 MT	RH 28022-PE105	Ø 25	↑	↑	718.5	↑	↑	↑		↑	
4		HE v1.8 AT	RH 28022-PE102	Ø 24	↑	↑	718.5	↑	↑	↑		↑	
5		HE v1.8 MT&AT	LH 28022-PE101	Ø 25	↑	↑	459.9	↑	↑	↑		↑	
6		HE v1.8 SUV 越野车	RH 28022-HB001	Ø 24	↑	↑	580.5	↑	↑	↑		↑	
7			LH 28022-HB002	Ø 22.5	↑	↑	455.2	↑	↑	↑		↑	
8		SC k1.4 MT	RH 28022-HB001	Ø 27	Ø 22	Ø 25.741	720	Ø 19.55	5	1.5		↑	
9			LH 28022-HB002	Ø 27	↑	↑	474.7	↑	↑	↑		↑	
10			LH 28022-HB003	Ø 27	↑	↑	480	↑	↑	↑		↑	
11			RH 28022-HB004	Ø 27	↑	↑	755	↑	↑	↑		↑	
12			LH 28022-HB005	Ø 27	↑	↑	445.7	↑	↑	↑		↑	
13	#24	VD v 2.0 AT	RH 28024-A7001	Ø 28	Ø 24	Ø 22	Ø 25.559	781.2	Ø 21.47	↑	1.8	↑	
14			LH 28024-A7002	Ø 28	↑	↑	470.4	↑	↑	↑		↑	
15	#25	CH455	LH 28025-C4011	Ø 28	Ø 25.4	Ø 28.918	581.1	Ø 22.52	↑	↑		↑	
16			RH 28025-C4012	Ø 27.5	↑	↑	485.8	↑	↑	↑		↑	
17			LH 28025-C4003	Ø 28	↑	↑	421.5	↑	↑	↑		↑	
18			RH 28025-C4004	Ø 28	↑	↑	589.5	↑	↑	↑		↑	
19			RH 28025-C4005	Ø 27.5	↑	↑	407.1	↑	↑	↑		↑	

References chart

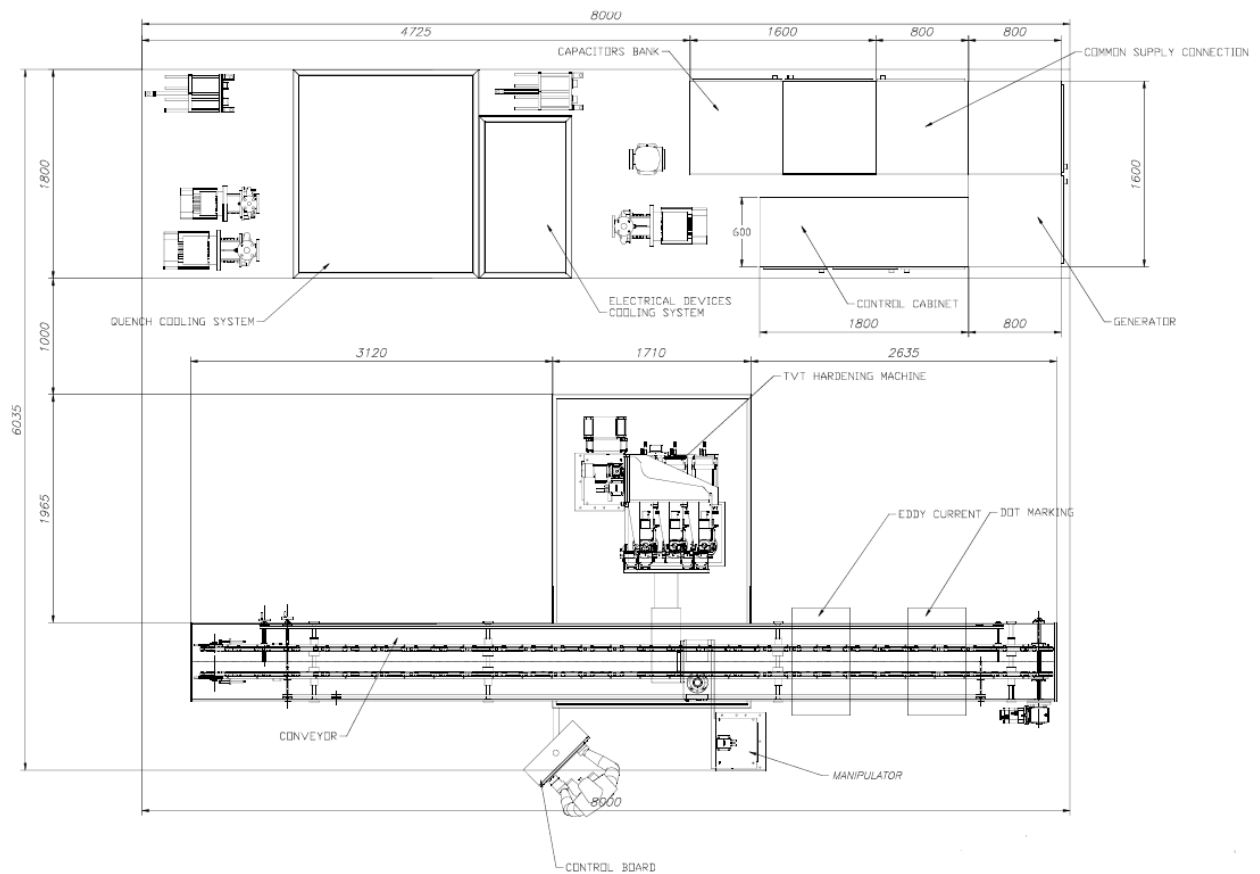
Material:	SAE10B38M2
Required cycle time:	23 sec average of RH and LH
Machine capability:	Max Ø 60 mm –Max Length 900 mm
Surface hardness:	55~59 HRc after hardening
Microstructure:	Pearlite + Ferrite
Deformations:	Serration O.P.D. max 0.02 mm
Lay-out:	Within L9000mm x W7000mm x H 4000mm
Remarks:	The validity of the present proposal is not extensive to other parts or applications not specified here.

1.3. Proposal

The whole installation includes the following circuit:

1. Conveyor.
2. Loading/Unloading gantry.
3. Hardening machine, vertical scanning 3 stations.
4. Eddy current and dot marking station.

For the hardening machine has been considered a multi output power supply of 300kW which allows an independent power per station of 100 kW to control and monitor independently every station at any moment.



Lay-out

1.4. Production conditions

Availability	
Cycle time	26" for 750 mm length part
Production rate	95%
Installation site	
Place, address	Monterrey (México)
Temperature, min. / max,	10-40°C
Relative humidity, max,	≤70%
Required supplies	
Electricity	3 x 480 V - 60 Hz
Industrial water	20 ÷ 28°C; 3 ÷ 7 bar
Compressed air	5 ÷ 6 bar

REMARKS:

- If the customer has not proper water utility circuit within his plant to fulfil required features, GH could provide consulting services, on the basis of your instructions, and propose alternative solutions.
- If the max inlet temperature is above 28°C, then a chiller unit with separate circuit for the converter is needed.

2. Scope of supply

The supplies related to the present proposal are:

2.1. Engineering		
Induction application parameters study and test	YES	
Equipment and project engineering	YES	
2.2. Manufacturing		
Components purchasing and equipment assembly	YES	
Assembly and put in operation	YES	
Technical documents	YES	
Pre-acceptance tests after manufacturing as itemized in § 8	YES	
Process statistic analysis & CPk ¹ verification before delivery		NO
2.3. Delivery		
Dismantling for transport and packaging	YES	
Transport and its assurances to final destination		NO
Handling in destination and placing in customer's factory		NO
Link to the customer's mains (AC supply, water, pneumatics...)		NO
Reassembly and put into production conditions on customer's plant	YES	
Training of customer's personnel	YES	
Final acceptance tests on customer's site	YES	
Technical support to the exploitation starting		OPTION
2.4. Standards		
Standard GH: Installations Manufacturing		
2.5. Technical documentation		
As itemized in Standard GH: Installations Manufacturing		
Documents	5 manuals (3 copies in Spanish + 2 copies in English)	

1. CPk: capability index intended to study the process repeatability and stability under normal work conditions

3. Equipment description

3.1. Conveyor

The parts are transported from the previous process (grooving) by a horizontal conveyor made by two parallel chains, step by step. The chains are placed over wear resistant guides.

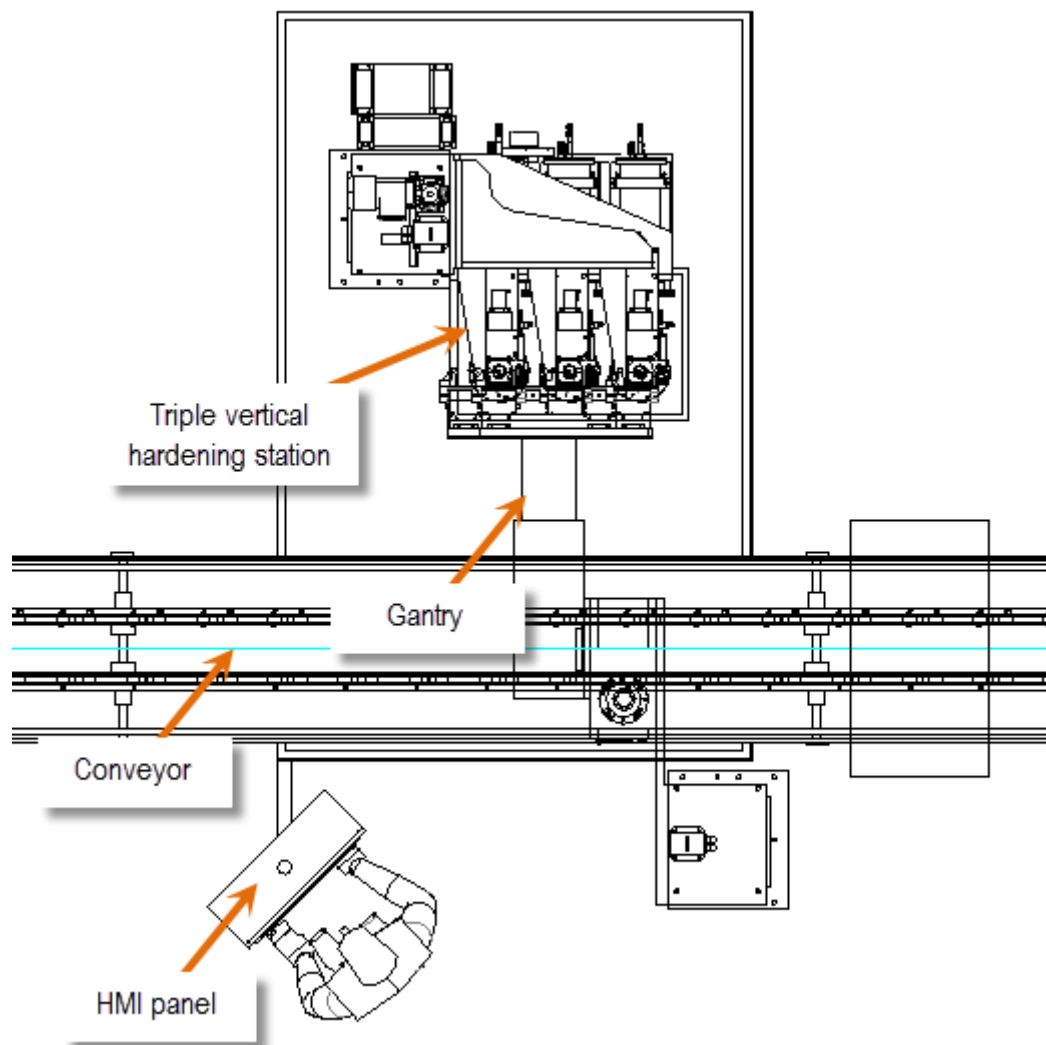
Once the parts are placed in the loading position, three parts are lifted by a pneumatic device. On this point they are axially referenced and checked on length and position.

Fig.1 Horizontal conveyor.



3.2. Hardening Machine

Fig.2 Gantry and hardening station.



3.2.1. Gantry

A gantry with 6 pneumatic clamps is in charge of the pickup of the parts to conveyor, loading and unloading of the parts in the hardening machine, and final put out on output conveyor.

The gantry has a servo-controlled vertical movement, indexing turning device and pneumatic for horizontal and longitudinal movement.

3.2.2. Hardening machine

This machine is able to treat three parts at a time which rotation is driven by a common motor.

The machine capacity is: part's length: 300...900 mm, diameter: 19...60 mm

In order to fix the part, a pneumatically driven tailstock automatically presses the part against the fixing tool. Furthermore, these tailstocks include a part rotation detection system (4 inputs per turn) as well as a position sensor which allows a part's length control. The lower tailstock is motorized to drive the turn to the parts.

This machine includes a dry/wet area separation which isolates the mechanical devices from the hardening area.

The elements in contact with hardening liquid are corrosion resistant materials, such as G10-for the isolating materials on the inductors. There is a physical separation between the dry zone and the wet area of the hardening station, so the components such as transformers, columns, ballscrews, etc. are safe of splashes of the hardening liquid as well as the vapors is minimized into this dry zone, preserving them and enlarging their life.

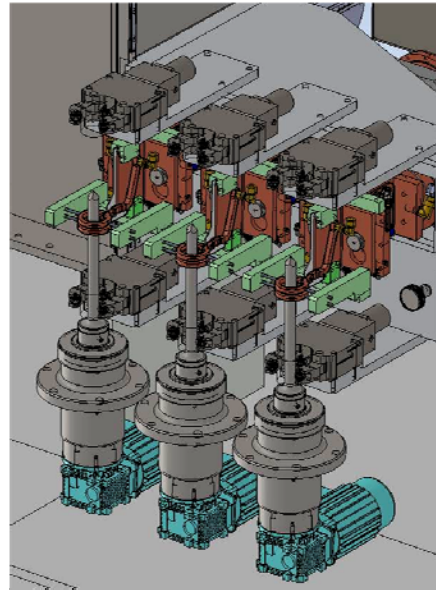


Fig.3 Vertical scanning machine.



3.2.3. Support rollers

To avoid the bending effect during the hardening two sets of 3 support rollers each has been included on supply.

These rollers are leaning the parts on the top and the bottom area of the shafts during the hardening avoiding deformations. Their horizontal movement are servo-controlled (independently the top set of rollers to the bottom) to have a complete control regulation during heating and each roller can be adjusted independently.

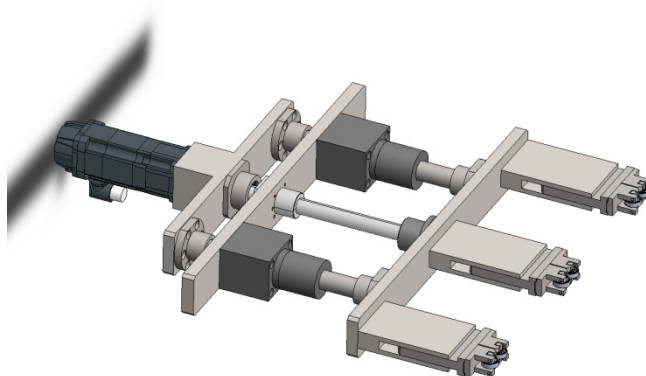


Fig.4 Support rollers set.

3.2.4. Bad and sample parts

NG parts and sample parts are left at the end of conveyor throw an especially dedicated ramp.

3.3. Eddy Current and Dot marking station

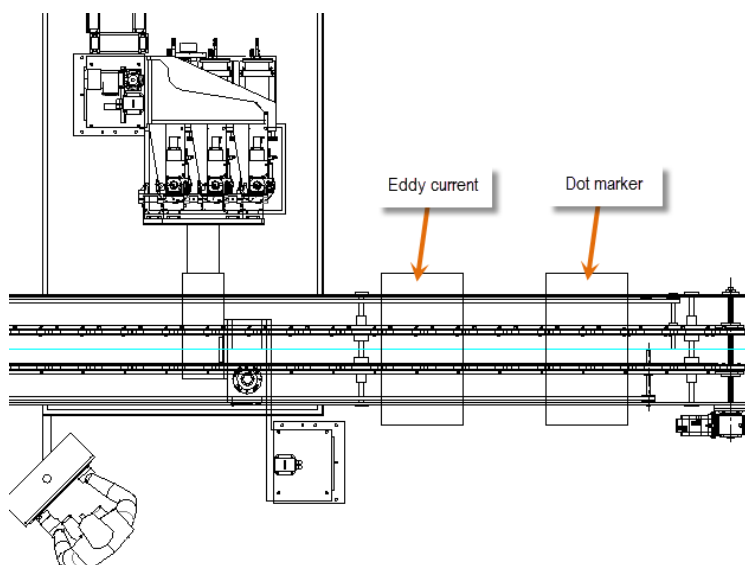


Fig.5 Eddy current and dot marker.

Eddy Current and dot marking are placed in two different stations.

In both cases the parts are raised pneumatically (one by one) and fixed by a clamp.

Eddy current is IBG: Eddy Visor DS Model following specifications.

Dot marker is JEIL MSTECH brand: Marksman MCU-100N Model following specifications.

*1 Note: Noise level close to the dot marking station won't accomplish <75dB noise level even inside an acoustic capsule isolation.

3.4. Electric system, control and command devices

1. Control system

The main function of the control & command system is the machine running and managing always subordinated to the safety of personnel and environment.

The hardening and tempering machines have independent control systems.

In the hardening machine is composed by:

Computer Numeric Control (CNC)..... Siemens 840 Dsl, PCU 50, with 5 axes servo controlled
(1 of them controlled by CNC)

Man - Machine Interface (MMI) Siemens OP12



Hand held - Machine Interface (HMI) Siemens HT2

2. Quality process supervision system

The operation concerned by this proposal is controlled by supervision of the following parameters:

MF/HF output frequency	YES	
MF/HF output power.....	YES	
Energy delivered during the heating	YES	
Heating time	YES	
Product cooling water flow	YES	
Product cooling water temperature	YES	
Quenching time	YES	
Product rotation	YES	
Product position and/or speed.....	YES	
Coil position and/or speed.....	YES	
PMS - GH Process Graphic Control	YES	
Datalogger	YES	

Parameters supervision:

- Integrated in control & command system (CNC o PLC)
- Performed by independent system

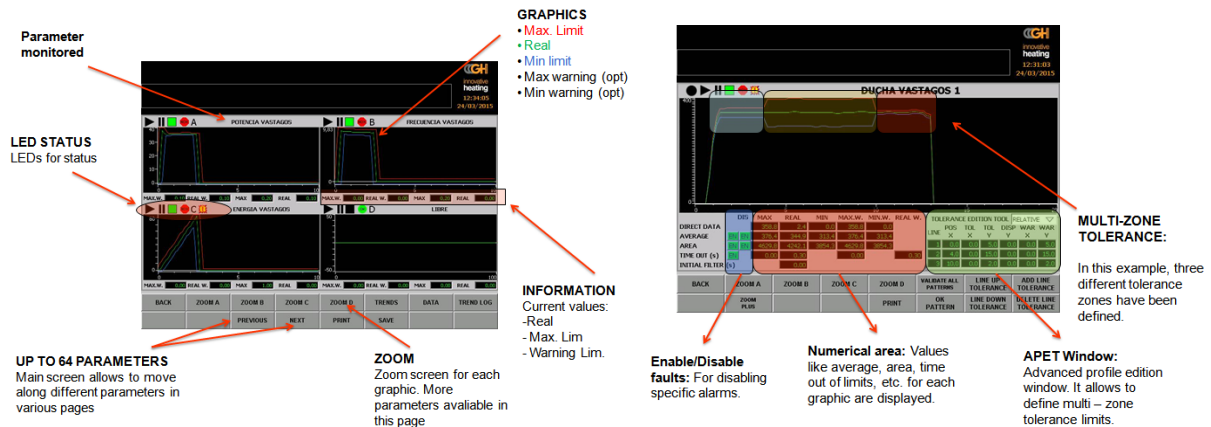
In case of non-conformity with the reference values of one or several parameters, the quality following system

Alerts the operator with an alarm signal	YES	
Starts the product automatic separation.....	YES	
Stops the machine.....	(User setting)	

Some screen examples of PMS - GH Process Graphic Control

GHPGC is the process monitoring system (PMS) developed by GH that allows to monitor and to control main heating process parameters in the safest and most functional way.

Fig.17 PMS screen examples.



It is carried out the graphic control of up to 99 parameters such as (*depending on the sensors and devices installed in machine):

Heating process	Cooling process	Hardening depth
Heating time	Quenching time	Frequency
Part temperature	Quenching flow	Heating time
Part energy	Quenching temperature	
Power	Quench polymer concentration	
Voltage		
Current		
Frequency		

In order to evaluate if a process is valid main parameters are recorded and plotted in real time according sampling time during whole cycle. When a good part is validated, MAX and MIN limits are calculated and plotted for each parameter in accordance to the tolerance values defined by the customer. After this moment, parts produced are monitored and compared with the defined limits. If any parameter detected is out of the limits, an alarm is triggered and part is considered as a NOK part.

3. Generalities

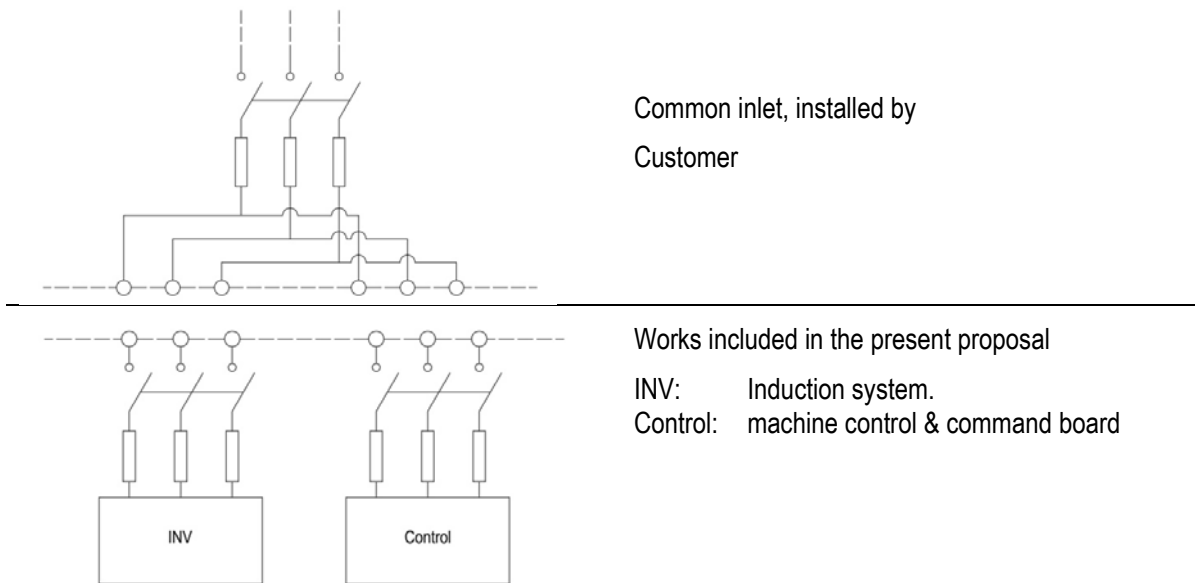
The electric system is made in accordance with standard GH NG 1.7

The execution of electric inlet includes two independent lines:

- One, for the induction equipment (generator and peripherals) and
- Other, for the general control system, mechanisms and electric devices in general.

This solution allow to keep a part working while the other is disconnected, functioning mode necessary for induction generator adjusting, maintenance and repairing operations.

This double inlet is compatible with an eventual common line, usually installed by the customer, with the usual protection, measure and/or breaking elements.



The protection grade of boards is:

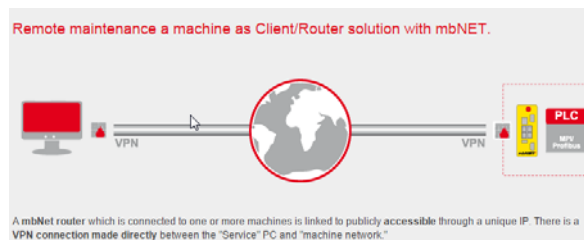
IP Material

54 Electric boards in general: generator, oscillating circuit, command panels

3.5. GH Remote assistance service

With the aim to assure a better service meanwhile a remote assistance service, GH installs a wireless router in the machine which allows a remote access to our technical service staff to solve possible incidences directly in the PLC from our own facilities in GH Electrotermia (Spain). This new tool would help us in a quicker answer or in case of any technical assistance.

The access to the machine is done throw a secure VPN connection by a dedicated hardware (WAN or 3G connection). Internet SIM card is not included on supply.



This service will be given free of charge during all the warranty period. After this time the customer may return the hardware to GH or buy it for 1.700,- €. The service fee for a 24h assistance during a year (after warranty period) will be of 1.000,- € and in case of technical assistance requirement the charge will be measured per hours (90€/hour).

3.6. Transistors induction generator (Series)

3.6.1. Functioning

The induction generator is the system destined to convert the energy coming from the industrial AC mains in a medium or high frequency signal adapted to the feeding of the induction heating station. This signal is a square wave that goes through an oscillating circuit (capacitors bank and an inductor as a minimum) and origins a magnetic field. Once that a part is under this magnetic field, electrical currents are made upon it. These currents increase the superficial temperature of the part by means of the Joule effect.

To obtain the square signal, the AC industrial current is rectified to a DC current. After this, a transistors inverter bridge obtains the desired AC wave at the adequate frequency for the application.

3.6.2. Security

The ensemble formed by generator and oscillating circuit is water refrigerated. The equipment protection is granted by the Reading of the integrated flow-meters. The control electronics keeps always protected the semiconductors against internal and external failures (inductor breakage, short-circuits, overloads...).

GH generators accomplish with Machine Security Standard ISO 13849-1, category PL d

3.6.3. Technical data

The selected generator is a MO3x100SM20 multiple output model being its power 100 kW for each station respectively. The following chart shows the main features of SM generator series.

Type		12SM	25SM	50SM	75SM	100SM	150SM	200SM	250SM	300SM	400SM	500SM	600SM	800SM	
Output continuous power	kW	12	25	50	75	100	150	200	250	300	400	500	600	800	
Frequency	kHz	[0.5, 20] [20, 150]						[0.5, 20]							
Power supply	kVA	15	31,25	62,5	93,75	125	187,5	250	312,5	375	500	625	750	1000	
Voltage supply		380–480 VAC 50/60 Hz				380-480 VAC 50/60 Hz ⁽¹⁾									
Generator width	in/mm	23,6/600				31,5/800		47,2/1200		55,1/1400		55,1/1400		102,3/2600	118,1/3000
Generator depth	in/mm	23,6-31,5/600-800				31,5/800		31,5/800		31,5/800		35,4/900		35,4/900	
Generator height	in/mm	43,3/1100				78,7/2000		78,7/2000		78,7/2000		78,7/2000		78,7/2000	
Width with cooling system	in/mm	23,6/600				78,7/2000		78,7/2000		86,6/2200		90,5/2300		137,8/3500 157,5/4000	
Depth with cooling system	in/mm	23,6/600		31,5/800		31,5/800		31,5/800		31,5/800		35,4/900		35,4/900	
Height with cooling system	in/mm	74,5/1900				78,7/2000		78,7/2000		78,7/2000		78,7/2000		78,7/2000	
Water temperature min/max	°F/°C	[68, 98,6] / [20, 37]													
Water supply		1"									1" ¼"		1" ½"		
Waterflow	gpm/lpm	2,6/10				3,9/15		7,9/30	8,8/40	12,7/48	18,5/70		30,38/115		36,7/140

3.6.4. Generator interface

The induction generator allows its control and command interface through an 8" panoramic colour touch screen (TouchHMI). Its user friendly interface makes able to supervise the equipment general functioning and to control the output signal.



Fig.18 Power supply screen examples:

- A: Main screen
- B: Flow-meters visualization
- C: Process parameters configuration
- D: Configuration screen

The interface allows the user to visualize the state of the equipment (prepared, heating, fail); visualize alarms (25 last events); configure the functioning parameters of the equipment. All this is done on an easy and intuitive way, "surfing" through different screens:

- Visualization of generator's analogical parameters: power, current, voltage, frequency and energy. Power regulation set.
- Screens related to the watch of flow-meters.
- Screens for parameters configuration and functioning mode.

Optionally, not included on the base supply, the generator can store programs to implement temperature controlled automatic heating cycles.

Also optionally, not included on the base supply, the generator can work as the main control for a simple machine by means of an internal program system (IPS) developed by GH. This program must be done by GH personnel.

3.6.5. Connection to an external control system

The generator is also ready to be controlled from an external system, so that the induction heating equipment is integrated into a system controlled by a general device. In order to do this, it could be linked through bus connections for its external control by means of: Profibus, Profinet, Device-net, Interbus.

Optionally, the generator can be equipped with different analogical inputs / outputs destined to exchange the following signals:

3.7. Oscillating circuit

The capacitors bank is the element providing the necessary capacity to establish, with the self-inductance of the inductor eventually augmented by the impedance adaptation transformer when applied, the resonant output circuit which will define the frequency of the induction signal.

Essentially, the oscillating circuit consists of one or several capacitors chosen depending on the oscillating conditions to state.

The capacitors are built in mixed dielectric, biodegradable, free of any pollution danger. The permissible overload is 1,35 the nominal current, including harmonics.

The oscillating circuit includes a coil interface transformer. This is a device intended to increase the coil self-inductance, when this value becomes too low to compose an oscillating circuit with an appropriated level of power allowing the induction heating.

The working principle of this transformer is based on the multiplication of the coil self-inductance, L , by a value, n , given by the transformer ratio.

Depending on the required ratio, the working voltage, power and frequency to apply, the transformer may be manufactured under different versions but it is always made under an isolating and protection cover.

Constructive solution details

Capacitors bank with adjustable capacity.....	YES
Oscillating circuit under IP54 cover	YES
Oscillating circuit equipped with coil interface transformer	YES
Hardening quench circuit equipped with quick connection plugs	YES
Coil interface transformer equipped with adjusting taps.....	YES

The safety against electric risks for access to zones under voltage is guaranteed by:

- Occasional access zones: panels screwed or fixed by other elements needing tools for dismantling.
- Doors: lock with key plus opening detection with automatic disconnection of the power

3.8. Coil or inductor

The coil or inductor is the element in the induction heating system performing the direct action on the product to process, it means: it is the component destined to transfer the energy to the material to treat. With this aim, the inductor is built with the form and size appropriated to the product.

The coil is manufactured as an assembly rigid and stable against the mechanical or thermal efforts, allowing a repetitive and controlled operation and a free of errors placing and positioning.

The main parts are:

Coil: with a turns number and shape specially chosen to obtain maximum efficiency. The coil is built with copper pipe intended to operate as internal cooling circuit destined to evacuate the residual heat due to the Joule effect losses.

Fastening system: it works as electrics connectors and as mechanical support and positioning.

Cooling circuit fitting: it is the water input / output elements connecting the coil to the coolant circuit.

The coil is equipped with a cooling quench destined to refrigerate the product after the action of the heating coil. This quench is built with magnetic inert materials in order to avoid any effect on inductor field.

Constructive solution details

Coil equipped with fast connecting system.....	YES
Coolant circuit equipped with fast connecting system	YES
Coil equipped with cooling quench	YES
Cooling quench equipped with fast connecting system	YES
Quench included in the machine	YES

After checking the given drawing for all the references GH Induction confirms we can accomplish the required cycle time with the same coil for all of them by scanning.

3.9. Induction heating cooling system

Some elements working in the induction heating system (e. g., transistors, capacitors, inductors) under the action of high currents require cooling to evacuate the heat by Joule effect.

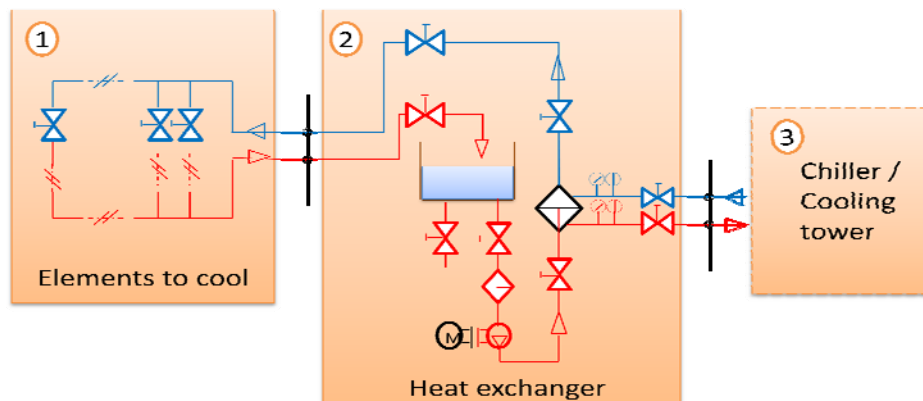
This cooling is based on a closed loop of liquid coolant through these elements. The system is named by GH as R3.

The liquid coolant applied in this circuit is demineralised² water. The low content of salts reduces the risk of obstruction by solids.

The coolant itself is cooled in a heat exchanger, where it transfers the heat to a second coolant, usually industrial water (water without any special chemical requirements).

Finally, the industrial water is cooled in a cooling tower or in a frigorific machine or chiller.

2. Avoid all confusion with distilled water which is not appropriated for this purpose because it acts as an acid due to its tendency to ionization under the temperature and voltage conditions present in the cooling circuits and in contact with the metallic parts of such circuits. The demineralized water is practically inert, dielectric and non-polluting (no risks are associated to the lose)



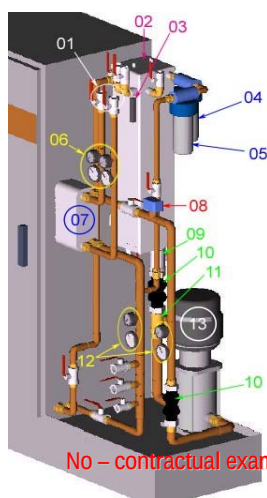
These three sections are independent systems:

1. The coolant circuit is integrated in the induction system and it is composed of a circuit with its appropriated control and measurement devices (valves, flow meters, thermostats...). Its function is to conduct the coolant through the components to cool in order to guarantee the induction system working temperature. In case of lack of capacity (due to coolant failure or excessive temperature), the integrated protections stop the equipment under failure condition.
2. The R3 unit is a system intended to maintain the coolant temperature as a thermal interface between the coolant circuit and the industrial water network
3. The external cooling equipment (usually a cooling tower or a chiller) is a part of the industrial water network not included in the induction heating equipment.

See in the chapter concerning the scope of supply the parts included in the present proposal.

3.9.1. R3 unit – demineralised water cooling system

The cooling R3 system is a well proven standard design of GH, fitted into most GH machines and generators. Treated water is used as coolant for the equipment in a closed circuit and is cooled at its turn by industrial water as secondary coolant in open circuit. The heat exchanger is of plate type.



No – contractual example

- 01 : Industrial water input/output
- 02 : Upper access to the cooling water tank
- 03 : Level indicator
- 04 : Polypropylene filter
- 05 : Deionising resin vase
- 06 : Thermometer & manometer
- 07 : Heat exchanger
- 09 : Output valve
- 10 : Attenuating connectors
- 12 : Thermometer & manometer
- 13 : Pump

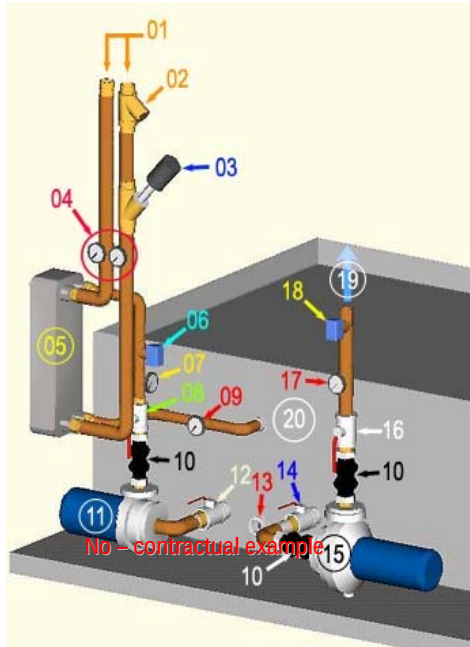
3.10. Hardening liquid cooling circuit

The product cooling water is applied on the product by quenching through a specific quench. The flow and temperature are measured in order to guarantee the process stability and repeatability.

A pump installed on the circuit origin keep the water circulation. After quenching, the water is recuperated by means of a recycling in a closed loop after conditioning. This system is named R4 by GH.

3.10.1. R4 unit - Description

The system is specially built to work with the usual product coolant as water containing soluble polymers additives. The excess of heat is taken out from the coolant by an industrial water flow through a heat exchanger. The standard construction and main components is:



- 01: Input / output industrial water connections
- 02: Y - type filter for protection of circuit against solid elements
- 03: Pneumatic switch valve (component applied eventually)
- 04: Manometer for primary circuit verification
- 05: Heat exchanger (remark: exchanger represented on the schema is standard welded type, depending of the specific conditions or under request, this exchanger may be dismountable type)
- 06: Product cooling water flow detector.
- 07: Product cooling water thermometer (exchanger inlet side).
- 08: Pump isolating valve (outlet side).
- 09: Product cooling water thermometer (exchanger outlet side).
- 10: Elastic connector for vibrations attenuation.
- 11: Product cooling water pump.
- 12: Pump isolating valve (inlet side).
- 13: Product cooling water tank outlet.
- 14: Quench isolating valve (inlet side).
- 15: Quenching pump.
- 16: Quench isolating valve (outlet side).
- 17: Product cooling water manometer (quench side).
- 18: Product cooling water flow detector (quench side).
- 19: Quench circuit.
- 20: Product cooling water tank.

4. Technical features

4.1. Induction hardening machine

4.1.1. Hardening solid state generator

Model.....	MO 3x100SM20
Output power.....	3x100 kW
Nominal frequency range	0.5-20 kHz
Input voltage.....	3x480V – 60 Hz
Technology	IGBT Series digital
MF / HF voltage.....	450V
Energetic efficiency	92%

4.1.2. Oscillating circuit capacitors bank

Working frequency	3 ~ 4 kHz
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4.1.3. Coils interface transformer/s

Model	Toroidal 100kW
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4.1.4. Coil / inductor

Type	Scanning hardening
Includes product cooling quench.....	YES
Water fastening type	Fast connector

MF/HF electric connection method Quick mechanical connection (Clamps)

4.1.5. Induction system coolant

Model R3
 Industrial water required temperature 22° C
 Industrial water required pressure 2...7 bar
 Heat exchanger type Welded
 Additional equipment Overpressure pump; Stainless steel piping

4.1.6. Product coolant

Model R4
 Industrial water required temperature 22° C
 Industrial water required pressure 2...7 bar
 Heat exchanger type Welded
 Additional equipment Paper filter; Stainless steel piping

4.1.7. Product handling system

Operation or stations: Hardening
 Model or type TVT-133
 Number of stations 3 scanning
 Functioning principle Vertical scanning
 Electric supply: 3 x 480 V – 60 Hz
 Pneumatic supply 5 ÷ 6 bar

4.1.8. Hardening machine control system

Computer Numeric Control (CNC): Siemens 840 D, PCU 50, 5 axes, 1 controlled by CNC
 Man - Machine Interface (MMI) Siemens OP12
 Human held unit (HMI) Siemens HT2

4.2. Painting

According to GH Standard, as follows:

Electrical cabinet (inside/outside) and raceways Grey RAL 7035
 Electrical panel Zinc coated
 Mechanical tooling (base and frame) Grey RAL 7035
 Pipes Green RAL 6018
 Machine inner tank Moist Metal Grip
 Type of painting PU Bi-component
 Commercial and standard components as per suppliers

Other colors or standards requirements may suppose additional charges on project.

5. Commercial offer

5.1. Delivery and guarantee terms

The present offer is in accordance with GH Electrotermia S.A. sales gen. terms & standard GH NG 1.7.

5.2. Acceptation tests

After manufacturing, the equipment here proposed will be tested. The tests will be performed in the GH Electrotermia S.A. factory in San Antonio de Benagéber (Valencia, Spain). GH will confirm to the customer the date of equipment availability for tests. The tests will consist on:

1. Static tests (supply verification)
 - 1.1. Concordance of manufacturing to the contract..... YES
 - 1.2. Accordance with safety and ergonomics YES
 - 1.3. Technical documents consistency YES
2. Dynamic tests (running verification)
 - 2.1. Inverter and oscillating circuit YES
 - 2.2. Induction equipment cooling system..... YES
 - 2.3. Product cooling system..... YES
 - 2.4. Handling system YES
 - 2.5. Control and command system YES
 - 2.6. Quality process supervision system..... YES
3. Treatment or process verification
 - 3.1. Treatment or process accordance with specifications YES
4. Production
 - 4.1. Cadence / speed of production..... YES
 - 4.2. Starting time..... YES
 - 4.3. Stability of running (H hours test) NO
 - 4.4. Change of product / reference time NO

A representative of the buyer must attend the tests together with GH personnel in order to sign the acceptance document. The client may give GH notice of renouncement and the tests will be undertaken without the presence of the client's representative. The reception documents will then be valid as an acceptance document.

This report authorizes the delivery, signed by a customer's representative and a GH's representative, will be stated with the declaration of conformity –and subsequently, the authorization for delivery- or the modifications

5.3. Guarantee

The guarantee period is for 1 year after FAC (however, for 2 years after FAC for parts related to NC - submission of warranty letter) and the supplier undertakes to repair or replace as quickly as possible any parts of the supplies which, before the expiry of the guarantee period are proved to be defective due to bad material, faulty design or poor workmanship. The supplier shall bear the costs of remedying the defective parts in its works. If the repair cannot be carried out in supplier's works, the customer shall bear the related costs to the extent exceeding the customary costs of transport, personnel, travelling, living, dismantling and reassembly of the defective parts.

If dispatch or taking over or erection is delayed due to reasons beyond supplier's control, the guarantee period shall end not later than 18 months after supplier's notification that the supplies are ready for dispatch.

The guarantee expires prematurely if the customer or a third party undertakes inappropriate modifications or repairs or if the customer, in case of a defect, does not immediately take appropriate steps to mitigate the damage and informs the supplier of such defect.

5.4. Delivery terms following Incoterms 2010

FOB Port: Valencia (Spain)

5.5. Payment terms

20 %	Payable upon presentation of A/Bond (by T/T without delay).
65%	Payable upon shipping documents (by irrevocable and confirmed L/C, opened in a first class opened at last 3 months before FOB estimated date).
15 %	Payable upon FAC issued by Hyundai-WIA and Warranty bond (by T/T without delay).

5.1. Delivery time

15th September 2017 FOB Valencia (with a grace period of 15 days).

A Hyundai-WIA parts delay for the trials in GH will correspond in a delay of the delivery time by the same period of time.

5.2. Validity of the proposal

The present proposal is valid for 7 days.

5.3. Prices 7002903.5

(Prices include not taxes).

Currency: EURO

Item	Designation
1	Induction hardening equipment, including:
1.1	One (1) induction hardening power supply MO3x100SM20 of 3 independent 100kW outputs, series type. Including cooling by water/air exchanger.
1.2	One (1) Capacitor bank
1.3	Three (3) Toroidal output transformers
2	Heat Treat handling machine
2.1	One (1) vertical hardening machine TVT 133 type as described on quotation with three parts heating station.
2.2	One (1) set of safety covers and protections
2.3	One (1) common platform (skid format)
3	Control & Command system
3.1	One (1) Control & command system based upon CNC Siemens 840d SL with control panel (OP12) and process parameters control and supervision and HT2 hand held unit.
3.2	One (1) PMS graphic parameter recorder
3.3	One (1) cooling exchanger water/air for electric components.
3.4	One (1) Data & statistics analysis & recording system (Data logger)
3.5	One (1) Teleservice system for remote assistance by means of VPS
4	Pre-acceptance tests at GH Induction facility
5	Delivery and Installation
5.1	Dismantling for transport and packaging
5.2	Final acceptance tests on customer's site
6	Documentation
6.1	Three (3) paper set and e-copy of documentation in Spanish and two (2) in English.

Price in EUROS, Item 1-6, Incoterm 2010: EXW (Valencia):

565.000,-€
(-Discount)
376.438,-€

COOLING

	Units	Unit price	TOTAL
7 Induction components equipment cooling system GH-R3 type with heat exchanger and overpressure pump for inductors and stainless steel piping.	1	19.000,-€	19.000,-€ 12.661,-€
8 Quench liquid circuit system GH-R4 type with heat exchangers, heating resistances, stainless steel tank and stainless steel piping. It includes control and command system and paper filter.	1	49.000,-€	49.000,-€ 32.651,-€
9 Magnetic separator on R4 circuit	1	9.000,-€	9.000,-€ 5.997,-€

ACCESSORIES

	Units	Unit price	TOTAL
10 Input/Output conveyor (incl. poka yoke, NOK parts)	1	130.000,-€	130.000,-€ 86.625,-€
11 Automatic handling gantry	1	120.000,-€	120.000,-€ 79.961,-€
12 Inductor and quench set in Scanning (valid for all references)	3	3.000,- €	9.000,-€ 5.997,-€
13 Spare inductor set in Scanning (valid for all references)	3	1.800,-€	5.400,-€ 3.598,-€
14 Tooling set for coil's check (valid for all references)	1	1.800,-€	1.800,-€ 1.199,-€
15 Eddy current and dot marking station	1	175.000,-€	175.000,-€ 116.610,-€
16 Spare parts set*	1	32.000,-€	32.000,-€ 11.667,-€*

*Spare parts set supply for the 3 machines would be valued on 35.000€. Another 34.000€ would be let in stock in GH Mexico facilities during all warranty for its use if required. After that period Hyundai WIA can purchase this set for this value.

SERVICES

17 Sea-worthy packing and freight FOB Valencia	1	13.000,-€ 8.661,-€
18 Start up, installation and training of staff Hardening machine	3 Weeks	30.000,-€ 19.990,-€
19 Service offer pack (Training pack and 1 month production assistance)	4 weeks	32.000,-€ * 0,-€

* Service offer will be offered free of cost in case of purchase of the three projects for Mexico

TOTAL Price Items 1-19 Incoterm 2010:FOB Valencia (Spain)	762.056,-€
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"Your details will become part of a database owned by GH ELECTROTERMIA, S.A. and processed for the purposes of CUSTOMER MANAGEMENT, MAINTENANCE, PAYMENTS, DISPATCH OF COMMERCIAL COMMUNICATIONS AND NEWSLETTERS.

You also expressly agree that GH ELECTROTERMIA, S.A. may use your personal details to send, by any means, including electronic means, commercial information and advertising on its products and services. If you do not want your details to be used for these advertising purposes, please tick this box ☐.

Customer details will be processed at all times pursuant to Spanish Organic Law 15/1999, of 13 December, on Personal Data Protection.

You may exercise your rights to access, rectify, block or erase the data by writing to the database manager at the following address: C/. Vereda Real s/n, 46184 San Antonio de Benageber (Valencia)."